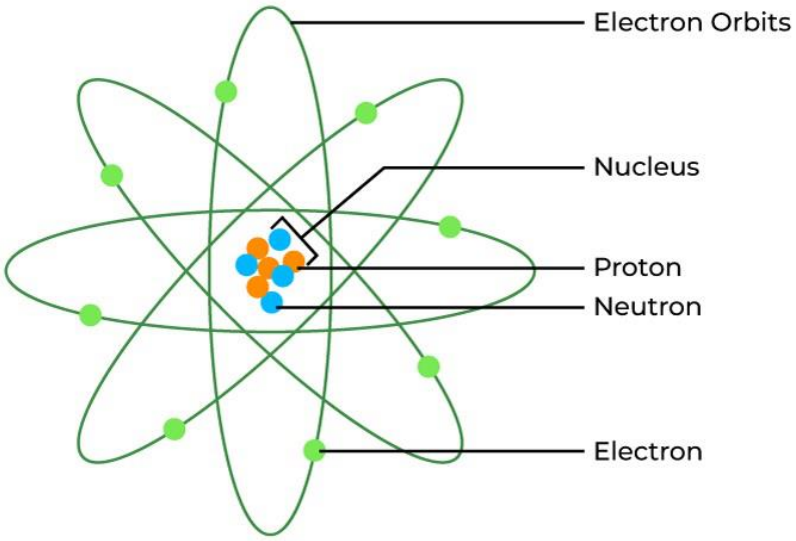
Fundamentals of Semiconductor

Q-1 Explain Structure of atom.



Electron

- Electrons are negatively charged particles which are moving around with high velocity around the nucleus of atom in specific path called orbit.
- The mass of electron is very small
- Electrons are revolving around the nucleus in a different orbit.
- Each orbit has certain fixed amount of energy.
- Electron denoted by ‘e’.

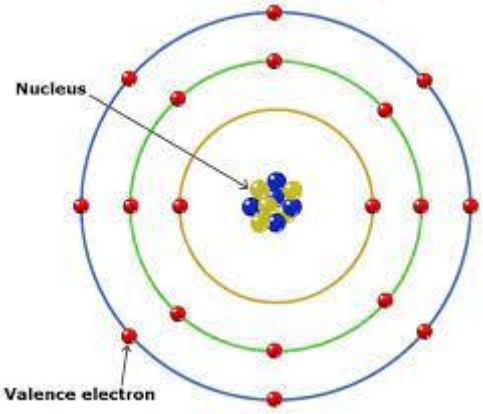
Proton

- Electrons are positively charged particles.
- It is heavier than electron.
- The charge of proton is equal to charge of electron but have opposite polarity.

Neutron

- Neutron has no charge.
- The mass of neutron is slightly higher than the proton.

Q-2 What is valance electrons



- The electrons are present in the outer most orbits or shell of an atom is called as valance electron.
- These electrons determine the electrical and physical properties of a material.
- The electrons revolving in the outer most orbit of an atom have highest energy level.
- Valance electrons are loosely attached to the nucleus.

1. When the number of valance electrons of an atom is less than 4, the material is usually known as a material. Example: Steel, Copper etc.

2. When the number of valance electrons of an atom is more than 4, the material is usually known as a material. Examples: Plastic, Rubber etc.

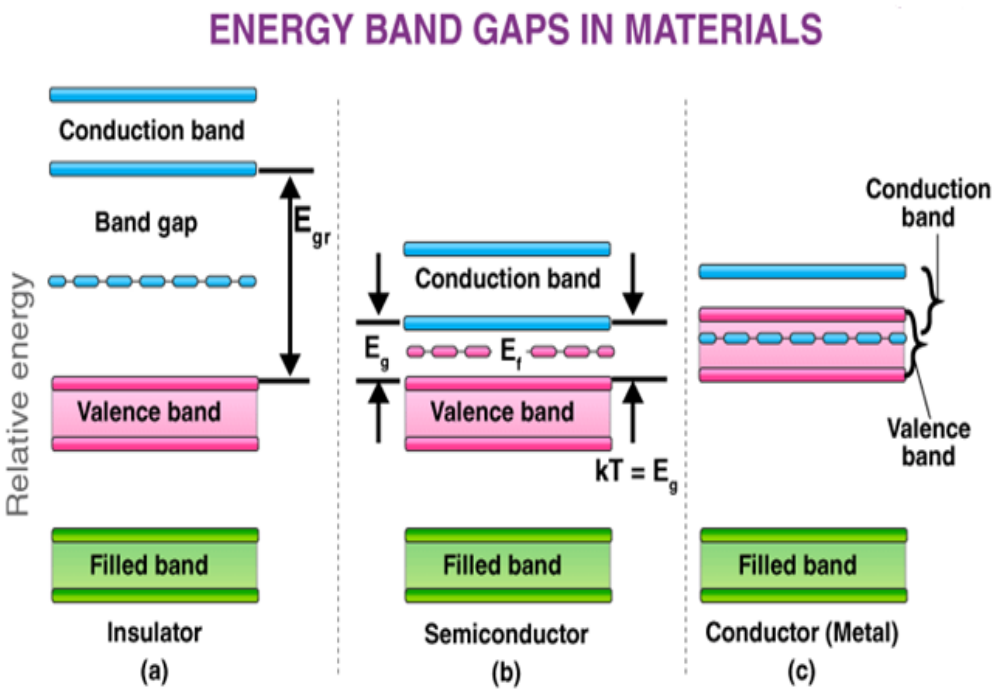
3. When the number of valance electrons of an atom is 4 , the ,material has both metal and insulator properties and usually known as a semiconductor. Example: Silicon, Germanium.

Q-3 What is Free Electrons

- The electron are not attached to the nucleus of a atom and free to move when external energy is applied are called free electrons.
- Valance electrons are loosely attached to the nucleus of a atom. When small amount of external energy form of heat or light is applied to the valance electrons they get pulled away from the parent atom and become free.
- A conductor is a substance which has large number of free electrons. When potential difference is applied across a conductor, the free electrons move towards the positive terminal of supply, constituting electric current.
- An insulator is substance which has practically no free electrons at ordinary temperature. Therefore an insulator does not conduct current under influence of potential difference.

- A semiconductor is a substance which has very few free electrons at room temperature consequently, under the influence of potential difference, a semiconductor practically conduct no current.

Q-4 Explain energy bands with diagram.



1. Valance Band:

- The electrons in the outermost shell are known as valence electrons.
- These valence electrons contain a series of energy levels and form an energy band known as the valence band.
- The valence band has the highest occupied energy.

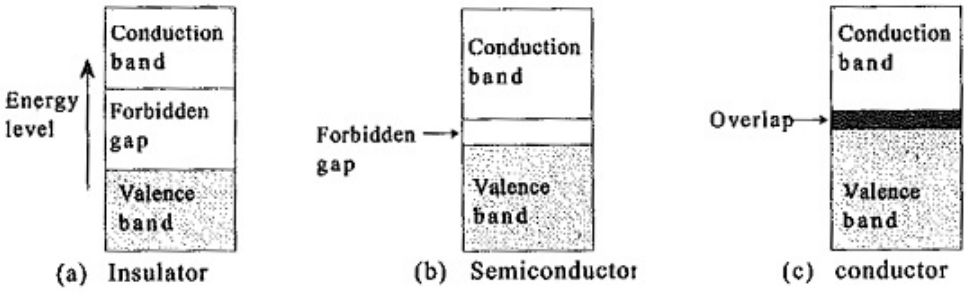
2. Conduction Band:

- The valence electrons are not tightly held to the nucleus due to which a few of these valence electrons leave the outermost orbit even at room temperature and become free electrons.
- The free electrons conduct current in conductors and are therefore known as conduction electrons.
- The conduction band is one that contains conduction electrons and has the lowest occupied energy levels.

3. Forbidden Band or Energy Gap:

- The gap between the valence band and the conduction band is referred to as the forbidden gap.
- As the name suggests, the forbidden gap doesn't have any energy and no electrons stay in this band.
- If the forbidden energy gap is greater, then the valence band electrons are tightly bound or firmly attached to the nucleus.
- We require some amount of external energy that is equal to the forbidden energy gap.

Q-5 Explain Conductor,Insulator and Semi conductor using their energy bands.



Energy band diagrams for insulators, semiconductors, and conductors.

1. Conductor

- At room temperature, some material has the large number of free electrons along with valance electrons. So it's does not require any external energy to make the free electrons available for the conduction. This material known as conductor.
- Example: Steel, Copper etc.

2. Insulator

- This material has a large band gap. Due to large band gap, the electrons from the valence band can't move into the conduction band. Hence the valence band remains to fill and the conduction band remains empty. This material known as insulator.
- Example: Plastic, Rubber etc.

3. Semi Conductor

- At room temperature, some material has very less number of free electrons. They require some external energy. If this much energy is supplied to the types of materials, the large number of free electrons are made available.
- This material does not behave fully as conductor or fully as insulator that type of material is known as a semi conductor.
- Example: Silicon, Germanium etc.

Q-6 Define Following term Doping, Intrinsic semi conductor and Extrinsic semi conductor.

1. Doping

The process of adding impurities in a semi conductor material is known as a doping.

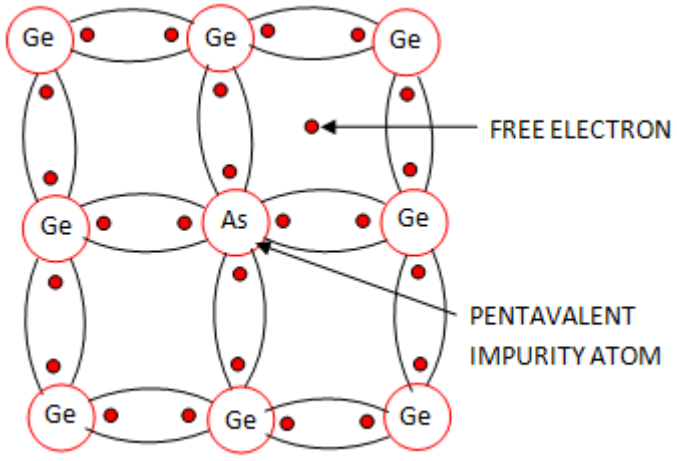
2. Intrinsic semi conductor

A semi conductor in an extremely pure form of semi conductor material is known as an Intrinsic semi conductor.

3. **Extrinsic semi conductor**

When adding a small amount of suitable impurities to a semi conductor. It is called impurity or Extrinsic semi conductor.

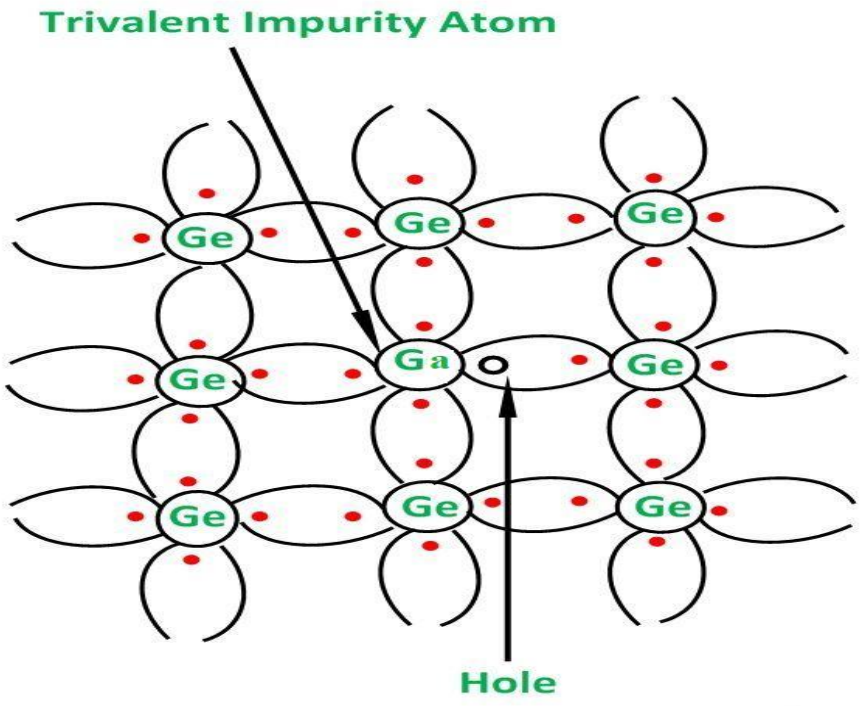
Q-7 Explain N- type semi conductor material with diagram.



N- Type semi conductor

- *When a small amount of pentavalent impurity is added to a pure semi conductor. It is known as* ***n-type semi conductor.***
- The five valance electrons impurity material is known as pentavalent atom.
- Above diagram show the formation of N-type semi conductor material by adding a arsenic material into a germanium.
- The arsenic material has the five valance electrons.
- In germanium semi conducting material, the arsenic fits in such a way that its four valance electron is set with four germanium atoms and fifth electron enters into the conduction band as free electron.
- It means that each arsenic atom gives one free electron when it is added into germanium atom. As the free electrons has the negative charge, such material known as N-type semi conductor.
- N-type semi conductor material is also known as ***donor*** impurities because they donate free electron to the semi conductor.

Q-8 Explain P-type semi conductor material with diagram.

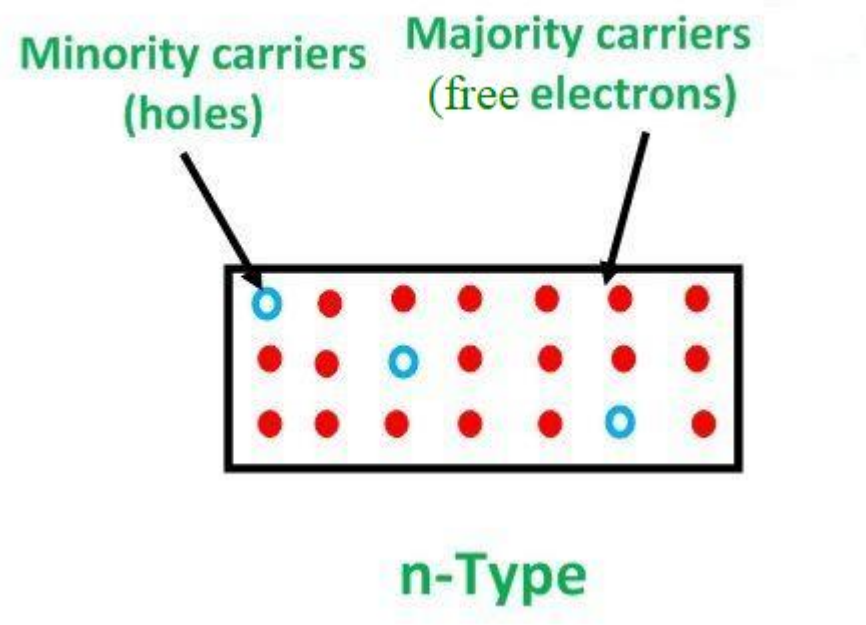


P-type semi conductor

- *When a small amount of trivalent impurity is added to a pure semi conductor. It is known as* ***P-type semi conductor.***
- The three valance electrons impurity material is known as *trivalent atom.*

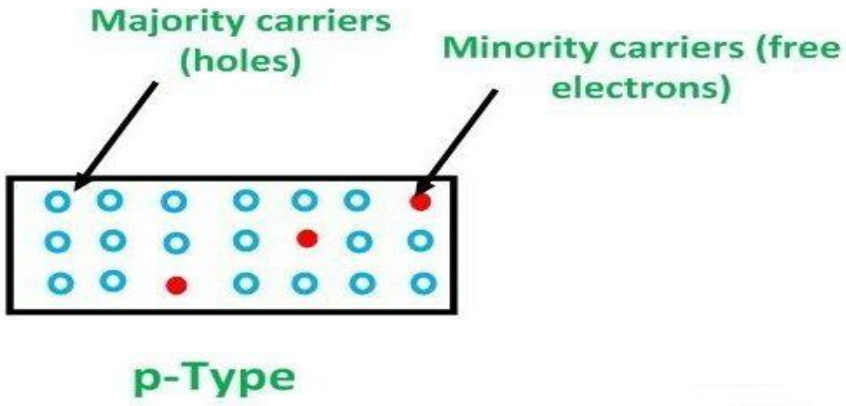
- Above diagram show the formation of P-type semi conductor material by adding a gallium material into a germanium.
- The gallium material has the three valance electrons.
- In germanium semi conducting material, the gallium fits in such a way that its three valance electron is set with four germanium atoms but due the short of the forth electron, the forth covalent band is incomplete. The remaining is called a Hole.
- P-type semi conductor material is also known as ***acceptor*** impurities because they accept hole to the semi conductor.

Q-9 Explain Majority and minority charge carrier for N-type semi conductor.



- N- Type material has a large number of free electrons.
- However, it may be recalled that even room temperature, some co-valent bonds break, thus releasing equal number of free electrons.
- An n-type material has its share of electron-hole pairs but in addition has a much larger quantity of free electrons due to the effect of impurity.
- These impurity caused free electrons are not associated with holes, consequently an n-type material has a large number of free electrons and a small number of holes as shown in a figure.
- The free electrons in this case are considered majority carriers and the holes are the minority carriers.

Q-10. Explain Majority and minority charge carrier for P-type semi conductor.



- P- Type material has a large number of Holes.
- However, it may be recalled that even room temperature, some co-valent bonds break, thus releasing equal number of holes electrons.
- A p-type material has its share of electron-hole pairs but in addition has a much larger quantity of holes due to the effect of impurity.
- These impurity caused free electrons are not associated with holes, consequently an p-type material has a large number of holes and a small number of free electrons as shown in a figure.
- The free electrons in this case are considered majority carriers and the holes are the minority carriers.